

OpenStax Calculus Volume 2

Section 3.3 Exercises: Trigonometric Substitution

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Simplify the following expressions by writing each one using a single trigonometric function.

1. $4 - 4\sin^2 \theta$
2. $9\sec^2 \theta - 9$
3. $a^2 + a^2 \tan^2 \theta$
4. $a^2 + a^2 \sinh^2 \theta$
5. $16 \cosh^2 \theta - 16$

Use the technique of completing the square to express each trinomial as the square of a binomial or the square of a binomial plus a constant.

6. $4x^2 - 4x + 1$
7. $2x^2 - 8x + 3$
8. $-x^2 - 2x + 4$

Integrate using the method of trigonometric substitution. Express the final answer in terms of the variable.

9. $\int \frac{dx}{\sqrt{4-x^2}}$
10. $\int \frac{dx}{\sqrt{x^2-a^2}}$
11. $\int \sqrt{4-x^2} dx$
12. $\int \frac{dx}{\sqrt{1+9x^2}}$
13. $\int \frac{x^2 dx}{\sqrt{1-x^2}}$
14. $\int \frac{dx}{x^2\sqrt{1-x^2}}$
15. $\int \frac{dx}{(1+x^2)^2}$
16. $\int \sqrt{x^2+9} dx$

17. $\int \frac{\sqrt{x^2-25}}{x} dx$
18. $\int \frac{\theta^3 d\theta}{\sqrt{9-\theta^2}}$
19. $\int \frac{dx}{\sqrt{x^6-x^2}}$
20. $\int \sqrt{x^6-x^8} dx$
21. $\int \frac{dx}{(1+x^2)^{3/2}}$
22. $\int \frac{dx}{(x^2-9)^{3/2}}$
23. $\int \frac{\sqrt{1+x^2} dx}{x}$
24. $\int \frac{x^2 dx}{\sqrt{x^2-1}}$
25. $\int \frac{x^2 dx}{x^2+4}$
26. $\int \frac{dx}{x^2\sqrt{x^2+1}}$
27. $\int \frac{x^2 dx}{\sqrt{1+x^2}}$
28. $\int (1-x^2)^{3/2} dx$

In the following exercises, use the substitutions $x = \sinh \theta$, $x = \cosh \theta$, or $x = \tanh \theta$. Express the final answers in terms of the variable x .

29. $\int \frac{dx}{\sqrt{x^2-1}}$
30. $\int \frac{dx}{x\sqrt{1-x^2}}$
31. $\int \sqrt{x^2-1} dx$
32. $\int \frac{\sqrt{x^2-1}}{x^2} dx$
33. $\int \frac{dx}{1-x^2}$
34. $\int \frac{\sqrt{1+x^2}}{x^2} dx$

Use the technique of completing the square to evaluate the following integrals.

35. $\int \frac{1}{x^2-6x} dx$
36. $\int \frac{1}{x^2+2x+1} dx$
37. $\int \frac{1}{\sqrt{-x^2+2x+8}} dx$
38. $\int \frac{1}{\sqrt{-x^2+10x}} dx$
39. $\int \frac{1}{\sqrt{x^2+4x-12}} dx$
40. Evaluate the integral without using calculus: $\int_{-3}^3 \sqrt{9-x^2} dx$.
41. Find the area enclosed by the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

42. Evaluate the integral $\int \frac{dx}{\sqrt{1-x^2}}$ using two different substitutions. First, let $x = \cos \theta$ and evaluate using trigonometric substitution. Second, let $x = \sin \theta$ and use trigonometric substitution. Are the answers the same?
43. Evaluate the integral $\int \frac{dx}{x\sqrt{x^2-1}}$ using the substitution $x = \sec \theta$. Next, evaluate the same integral using the substitution $x = \csc \theta$. Show that the results are equivalent.
44. Evaluate the integral $\int \frac{x}{x^2+1} dx$ using the form $\int \frac{1}{u} du$. Next, evaluate the same integral using $x = \tan \theta$. Are the results the same?
45. State the method of integration you would use to evaluate the integral $\int x\sqrt{x^2+1} dx$. Why did you choose this method?
46. State the method of integration you would use to evaluate the integral $\int x^2\sqrt{x^2-1} dx$. Why did you choose this method?
47. Evaluate $\int_{-1}^1 \frac{x dx}{x^2+1}$
48. Find the length of the arc of the curve over the specified interval: $y = \ln x$, $[1, 5]$. Round the answer to three decimal places.
49. Find the surface area of the solid generated by revolving the region bounded by the graphs of $y = x^2$, $y = 0$, $x = 0$, and $x = \sqrt{2}$ about the x -axis. (Round the answer to three decimal places.)
50. The region bounded by the graph of $f(x) = \frac{1}{1+x^2}$ and the x -axis between $x = 0$ and $x = 1$ is revolved about the x -axis. Find the volume of the solid that is generated.

Solve the initial-value problem for y as a function of x .

51. $(x^2 + 36)\frac{dy}{dx} = 1$, $y(6) = 0$
52. $(64 - x^2)\frac{dy}{dx} = 1$, $y(0) = 3$
53. Find the area bounded by $y = \frac{2}{\sqrt{64-4x^2}}$, $x = 0$, $y = 0$, and $x = 2$.
54. An oil storage tank can be described as the volume generated by revolving the area bounded by $y = \frac{16}{\sqrt{64+x^2}}$, $x = 0$, $y = 0$, and $x = 2$ about the x -axis. Find the volume of the tank (in cubic meters).
55. During each cycle, the velocity v (in feet per second) of a robotic welding device is given by $v = 2t - \frac{14}{4+t^2}$, where t is time in seconds. Find the expression for the displacement s (in feet) as a function of t if $s = 0$ when $t = 0$.
56. Find the length of the curve $y = \sqrt{16-x^2}$ between $x = 0$ and $x = 2$.